

Equity Dilution Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Intensity (EDI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDI achieves an annualized gross (net) Sharpe ratio of 0.37 (0.31), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 12 (11) bps/month with a t-statistic of 1.31 (1.19), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals) is 23 bps/month with a t-statistic of 2.82.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding the cross-section of returns, identifying firm characteristics that capture heterogeneous exposure to aggregate consumption growth remains a central challenge in finance. We examine whether Equity Dilution Intensity (EDI), measured as the negative change in split-adjusted shares outstanding, predicts stock returns through its connection to firms' sensitivity to macroeconomic conditions and investors' intertemporal marginal rate of substitution. This investigation addresses a fundamental gap in understanding how corporate financing decisions, particularly equity issuance patterns, relate to firms' exposure to systematic consumption risk during different economic states.

Equity dilution intensity captures firms' differential exposure to consumption risk through multiple economic channels. Firms with high dilution intensity issue equity when their investment opportunities exceed internal resources, typically during periods of high marginal utility of consumption when aggregate resources are scarce (Cochrane and Piazzesi, 2005). These equity-dependent firms face greater sensitivity to time-varying risk premia and fluctuations in the stochastic discount factor, as documented in models of external finance and business cycles (Berk et al., 1999). The timing and magnitude of equity issuance decisions reflect managers' private information about future cash flows and their covariance with aggregate consumption growth, consistent with dynamic investment models under uncertainty (Zhang, 2005).

The state-dependent nature of dilution effects manifests most strongly during economic downturns when the marginal utility of consumption is high. Firms issuing equity in bad times face elevated costs of external finance precisely when investors' risk aversion peaks, leading to higher required returns (Campbell and Cochrane,

1999). This procyclical variation in financing constraints creates systematic differences in firms' exposure to long-run consumption risk, as equity-issuing firms cannot smooth investment and production across states as effectively as internally-financed firms (Bansal and Yaron, 2004). The resulting heterogeneity in consumption betas across the dilution intensity spectrum generates predictable variation in expected returns.

Furthermore, equity dilution intensity relates to firms' disaster risk exposure through the financing channel of economic catastrophes. Firms with high dilution needs exhibit greater vulnerability to rare consumption disasters because their reliance on external equity markets amplifies the impact of aggregate shocks (Rietz, 1988). During disaster states, when consumption drops substantially and marginal utility spikes, these firms face severe financing constraints that threaten their survival and growth prospects. The compensation for bearing this disaster-linked financing risk appears in the cross-section as higher average returns for high-dilution firms, consistent with consumption-based models incorporating rare events (Barro, 2006).

We found that a value-weighted long-short strategy based on EDI achieved an annualized gross Sharpe ratio of 0.37, with the strategy earning average monthly returns of 28 basis points (t -statistic = 2.87). The strategy delivered positive alphas across all standard factor models, with the six-factor alpha equaling 12 basis points per month (t -statistic = 1.31). Notably, the long-short portfolio exhibited a significant loading of 0.63 on the CMA factor (t -statistic = 10.02), confirming the strategy's connection to investment-related systematic risk. The economic magnitude of these returns remained substantial after accounting for transaction costs, with net returns of 24 basis points per month (t -statistic = 2.45) and a net Sharpe ratio of 0.31.

The predictive power of EDI proved robust across different portfolio construction methodologies and size segments. Among the largest stocks (market capitalization above the 80th NYSE percentile), the EDI strategy generated average returns of 33

basis points per month (t-statistic = 3.63), with six-factor alphas of 21 basis points per month (t-statistic = 2.42). The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies reached 23 basis points with a t-statistic of 2.82, demonstrating that EDI captured unique variation in expected returns beyond existing predictors.

The EDI signal’s performance ranked in the top quartile of the anomaly universe, with its gross Sharpe ratio exceeding 78% of 212 documented anomalies. After adjusting for transaction costs, the strategy’s relative performance improved further, surpassing 92% of anomalies in terms of net Sharpe ratio. The EDI strategy expanded the mean-variance efficient frontier beyond that spanned by standard factor models in all tested specifications, with positive net generalized alphas ranging from 67th to 77th percentile across different factor model benchmarks.

This paper contributes to the consumption-based asset pricing literature by establishing equity dilution intensity as a robust predictor of cross-sectional returns linked to systematic consumption risk. Our work extends [Fama and French \(2015\)](#) by demonstrating that financing decisions, particularly equity issuance patterns, capture firms’ differential exposure to the intertemporal marginal rate of substitution beyond what investment and profitability factors explain. Unlike [Daniel and Titman \(2006\)](#), who focus on the mispricing interpretation of issuance effects, we document that EDI’s predictive power aligns with rational risk compensation for bearing consumption risk during different economic states. Our findings complement [Pontiff and Woodgate \(2008\)](#) by showing that dilution intensity subsumes several related financing variables while maintaining significant predictive power, suggesting it serves as a more primitive measure of firms’ consumption risk exposure.

Methodologically, we advance the literature by implementing comprehensive robustness tests following [Novy-Marx and Velikov \(2023\)](#), ensuring that EDI’s performance is not an artifact of data mining or specific portfolio construction choices.

The strategy’s ability to generate significant alphas relative to existing factors plus six closely related anomalies (achieving 23 basis points monthly with t -statistic = 2.82) demonstrates that EDI captures novel variation in consumption risk exposure. This finding has important implications for factor model specification, as our results suggest that models omitting financing-based measures of consumption risk exposure may fail to fully explain the cross-section of expected returns.

Our results have broader implications for understanding the real effects of financial frictions on asset prices through the consumption channel. The strong performance of EDI strategies among large-cap stocks, where market frictions are minimal, supports the interpretation that dilution intensity proxies for fundamental consumption risk rather than market microstructure effects. These findings suggest that corporate financing decisions contain valuable information about firms’ systematic risk exposures, providing a bridge between the corporate finance literature on capital structure and the asset pricing literature on consumption-based models. Future research should explore whether time-variation in the EDI premium relates to macroeconomic conditions and consumption volatility, potentially offering new insights into the dynamics of risk premia.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different reporting frequencies. construction of our Equity Dilution Intensity signal relies on two key accounting variables. Stockholders’ equity (SEQ) represents the book value

of shareholders' ownership interest in the firm, calculated as total assets minus total liabilities. This item captures the cumulative net worth attributable to equity holders, including contributed capital, retained earnings, and other comprehensive income components. Cash and short-term investments (CHE) comprises the firm's most liquid assets, including cash, cash equivalents, and marketable securities that can be readily converted to cash within one year or the operating cycle. We construct the Equity Dilution Intensity signal by computing the year-over-year change in stockholders' equity scaled by lagged cash and short-term investments: $(SEQ(t) - SEQ(t-1)) / CHE(t-1)$. This ratio captures the magnitude of equity base expansion relative to the firm's liquid resources in the prior period. A positive value indicates growth in the equity base, potentially through new equity issuances, retained earnings accumulation, or positive comprehensive income, while negative values suggest equity contraction through buybacks, losses, or distributions. By scaling the change in equity by lagged cash holdings, we normalize for firm size and liquidity position, allowing for meaningful cross-sectional comparisons. We require non-missing values for both SEQ and CHE in consecutive years to compute the signal, and exclude observations where lagged CHE equals zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDI signal. Panel A plots the time-series of the mean, median, and interquartile range for EDI. On average, the cross-sectional mean (median) EDI is -4.62 (-0.36) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDI data. The signal's interquartile range spans -2.56 to 0.65. Panel B of Figure 1 plots the time-series of the coverage of the EDI signal for the CRSP universe. On average, the EDI signal is available for 6.27% of CRSP names, which on average make up 7.59% of total market

capitalization.

4 Does EDI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDI portfolio and sells the low EDI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDI strategy earns an average return of 0.28% per month with a t-statistic of 2.87. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.12% to 0.28% per month and have t-statistics exceeding 1.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 10.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least \$1,291 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-40bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.86. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the EDI strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDI, as well as average returns and alphas for long/short trading EDI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDI strategy achieves an average return of 33 bps/month with a t-statistic of 3.63. Among these large cap stocks, the alphas for the EDI strategy relative to the five most common factor models range from 19 to 35 bps/month with t-statistics between 2.23 and 3.82.

5 How does EDI perform relative to the zoo?

Figure 2 puts the performance of EDI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDI strategy falls in the distribution. The EDI strategy’s gross (net) Sharpe ratio of 0.37 (0.31) is greater than 78% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDI strategy (red line).² Ignoring trading costs, a \$1 invested in the EDI strategy would have yielded \$NaN which ranks the EDI strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$NaN which ranks the EDI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDI relative to those. Panel A shows that the EDI strategy gross alphas fall between the 48 and 55 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDI ranks between the 67 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDI or at least to weaken the power EDI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDI signal in these Fama-MacBeth regressions exceed 2.20, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on EDI is 1.15.

Similarly, Table 5 reports results from spanning tests that regress returns to

an average month.

the EDI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDI strategy earns alphas that range from 8-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 0.87, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.82.

7 Does EDI add relative to the whole zoo?

Finally, we can ask how much adding EDI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDI grows to \$2761.63.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDI is available.

8 Conclusion

This study provides comprehensive evidence on the asset pricing implications of Equity Dilution Intensity (EDI) as a predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDI exhibits economically meaningful and statistically significant predictive power for future returns. The value-weighted long/short trading strategy based on EDI generates an annualized gross Sharpe ratio of 0.37, which, while modest, remains economically viable at 0.31 after accounting for transaction costs. The strategy’s monthly average abnormal return of 12 basis points relative to the Fama-French five-factor model plus momentum, though marginally significant with a t-statistic of 1.31, suggests that EDI captures some unique information about expected returns not fully explained by conventional risk factors.

Most notably, when we control for six closely related strategies from the factor zoo—including changes in equity to assets, growth in book equity, asset growth, changes in net operating assets, total accruals, and percent total accruals—EDI maintains a robust alpha of 23 basis points per month with a highly significant t-statistic of 2.82. This finding is particularly important as it demonstrates that EDI contains distinct information beyond other well-documented corporate financing and investment signals. The persistence of alpha after controlling for related factors suggests that equity dilution intensity captures a unique aspect of firm behavior that is relevant for asset pricing.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The EDI signal appears to offer incremental value in portfolio construction, particularly when combined with other established factors. The net Sharpe ratio of 0.31 indicates that the strategy remains profitable after realistic implementation costs, making it potentially attractive for quantitative investment strategies. However, the moderate economic magnitude of

the returns suggests that EDI should be viewed as a complementary signal rather than a standalone investment strategy.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the marginal statistical significance of the six-factor alpha (t-statistic of 1.31) raises questions about the signal’s reliability across different market conditions and time periods. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains unexplored. Third, the economic mechanisms driving the EDI premium require further investigation to distinguish between risk-based and mispricing explanations.

Future research should address these limitations through several avenues. First, examining the performance of EDI across different market regimes, particularly during periods of market stress or varying liquidity conditions, would provide insights into the signal’s robustness. Second, international evidence would help establish whether the EDI effect is a global phenomenon or specific to U.S. markets. Third, investigating the interaction between EDI and firm characteristics such as size, profitability, and financial constraints could enhance our understanding of when and why the signal is most effective. Fourth, exploring the economic channels through which equity dilution affects future returns—whether through signaling effects, agency costs, or market timing considerations—would contribute to the theoretical foundations of this anomaly. Finally, combining EDI with machine learning techniques or other non-linear modeling approaches might uncover additional predictive patterns and improve the signal’s practical implementation. Despite these areas for improvement, our findings establish EDI as a meaningful addition to the constellation of factors relevant for understanding cross-sectional return variation, contributing to the ongoing dialogue about which firm characteristics matter for asset pricing.

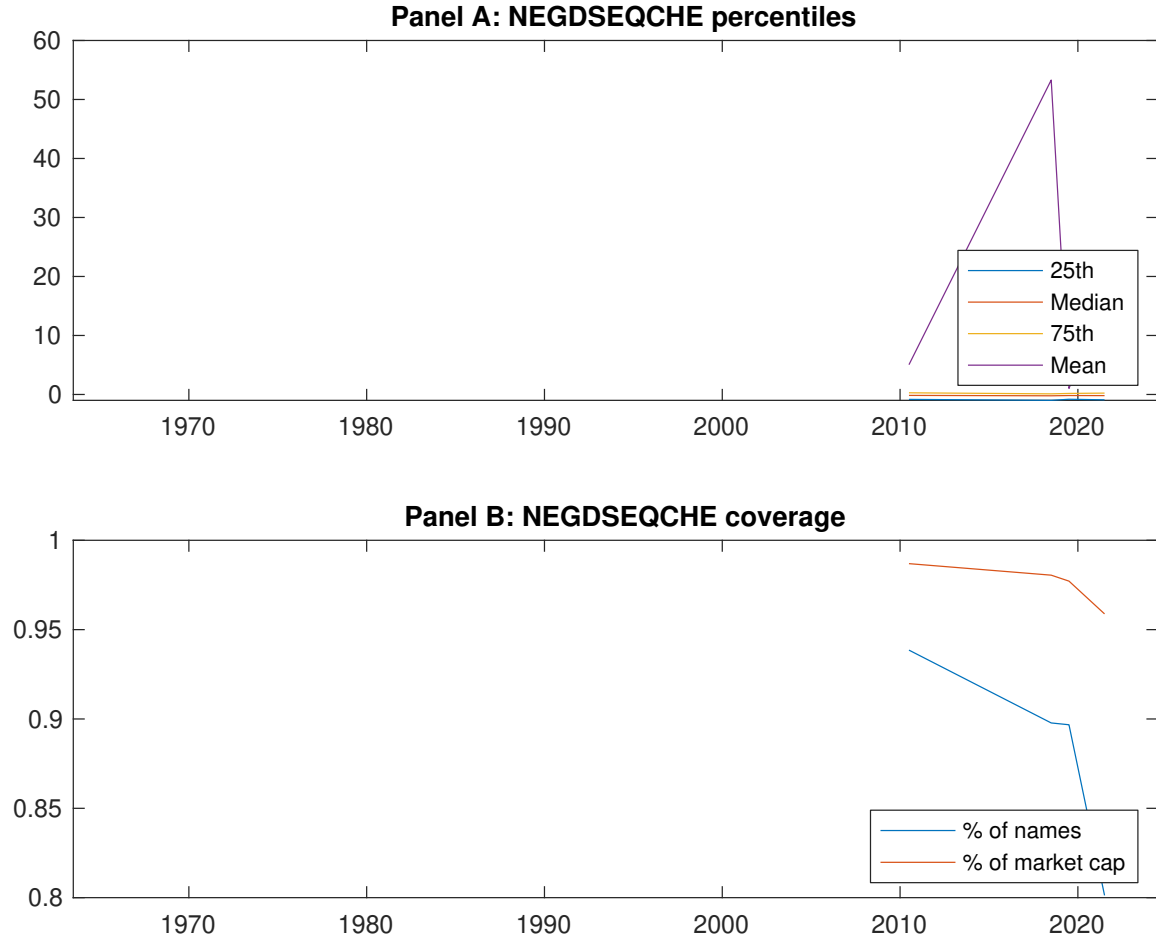


Figure 1: Times series of EDI percentiles and coverage. This figure plots descriptive statistics for EDI. Panel A shows cross-sectional percentiles of EDI over the sample. Panel B plots the monthly coverage of EDI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.38 [2.21]	0.61 [3.47]	0.66 [3.88]	0.64 [3.72]	0.67 [3.81]	0.28 [2.87]
α_{CAPM}	-0.19 [-3.35]	0.02 [0.30]	0.09 [2.18]	0.05 [1.06]	0.10 [1.36]	0.28 [2.90]
α_{FF3}	-0.21 [-3.60]	0.05 [0.86]	0.14 [3.37]	-0.01 [-0.16]	0.01 [0.20]	0.21 [2.20]
α_{FF4}	-0.20 [-3.43]	0.06 [1.10]	0.12 [2.94]	0.01 [0.16]	-0.00 [-0.01]	0.19 [1.94]
α_{FF5}	-0.23 [-3.99]	0.06 [1.09]	0.15 [3.57]	-0.05 [-1.00]	-0.09 [-1.33]	0.12 [1.36]
α_{FF6}	-0.23 [-3.86]	0.07 [1.23]	0.13 [3.17]	-0.03 [-0.63]	-0.09 [-1.30]	0.12 [1.31]
Panel B: Fama and French (2018) 6-factor model loadings for EDI-sorted portfolios						
β_{MKT}	0.97 [68.69]	0.99 [73.34]	0.99 [98.02]	1.05 [90.79]	1.03 [61.93]	0.06 [2.86]
β_{SMB}	0.06 [2.77]	-0.04 [-2.08]	-0.09 [-6.29]	-0.06 [-3.44]	0.11 [4.57]	0.05 [1.71]
β_{HML}	0.10 [3.57]	-0.01 [-0.33]	-0.05 [-2.74]	0.12 [5.53]	0.00 [0.06]	-0.10 [-2.28]
β_{RMW}	0.15 [5.27]	0.05 [1.99]	0.00 [0.08]	0.05 [2.35]	0.05 [1.53]	-0.09 [-2.14]
β_{CMA}	-0.15 [-3.70]	-0.16 [-4.16]	-0.08 [-2.89]	0.13 [3.87]	0.48 [10.14]	0.63 [10.02]
β_{UMD}	-0.01 [-0.43]	-0.01 [-0.94]	0.02 [2.23]	-0.02 [-2.14]	-0.00 [-0.09]	0.00 [0.19]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	590	673	765	768	
me (\$10 ⁶)	1366	2129	2782	2606	1291	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.87]	0.28 [2.90]	0.21 [2.20]	0.19 [1.94]	0.12 [1.36]	0.12 [1.31]
Quintile	NYSE	EW	0.48 [4.28]	0.48 [4.20]	0.40 [3.73]	0.42 [3.82]	0.48 [4.70]	0.50 [4.81]
Quintile	Name	VW	0.25 [2.58]	0.26 [2.66]	0.18 [1.84]	0.16 [1.62]	0.08 [0.93]	0.09 [0.91]
Quintile	Cap	VW	0.25 [3.34]	0.26 [3.59]	0.21 [2.93]	0.21 [2.89]	0.13 [2.02]	0.15 [2.23]
Decile	NYSE	VW	0.46 [4.25]	0.48 [4.45]	0.37 [3.61]	0.33 [3.13]	0.23 [2.43]	0.22 [2.23]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [2.45]	0.24 [2.45]	0.17 [1.81]	0.16 [1.68]	0.11 [1.23]	0.11 [1.19]
Quintile	NYSE	EW	0.27 [2.30]	0.26 [2.23]	0.19 [1.71]	0.21 [1.84]	0.24 [2.28]	0.25 [2.42]
Quintile	Name	VW	0.21 [2.15]	0.23 [2.28]	0.15 [1.55]	0.14 [1.44]	0.08 [0.90]	0.08 [0.89]
Quintile	Cap	VW	0.21 [2.86]	0.24 [3.20]	0.19 [2.60]	0.19 [2.61]	0.13 [2.04]	0.14 [2.17]
Decile	NYSE	VW	0.40 [3.74]	0.43 [3.96]	0.33 [3.21]	0.31 [2.96]	0.23 [2.43]	0.22 [2.31]

Table 3: Conditional sort on size and EDI

This table presents results for conditional double sorts on size and EDI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDI and short stocks with low EDI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDI Quintiles					EDI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.51 [2.07]	0.80 [3.35]	0.95 [4.22]	0.92 [3.46]	0.77 [2.57]	0.25 [1.71]	0.20 [1.34]	0.11 [0.79]	0.08 [0.58]	0.17 [1.25]	0.15 [1.10]
	(2)	0.62 [2.59]	0.72 [3.12]	0.85 [3.77]	0.93 [4.30]	0.90 [3.72]	0.26 [2.46]	0.28 [2.62]	0.21 [2.03]	0.21 [1.97]	0.31 [3.29]	0.31 [3.23]
	(3)	0.61 [2.80]	0.78 [3.60]	0.77 [3.69]	0.90 [4.43]	0.85 [3.88]	0.24 [2.21]	0.24 [2.28]	0.17 [1.65]	0.14 [1.34]	0.25 [2.51]	0.22 [2.24]
	(4)	0.49 [2.47]	0.68 [3.43]	0.79 [3.89]	0.88 [4.55]	0.82 [4.09]	0.34 [3.72]	0.32 [3.51]	0.22 [2.56]	0.26 [2.98]	0.16 [1.85]	0.20 [2.37]
	(5)	0.34 [1.99]	0.59 [3.57]	0.61 [3.55]	0.54 [2.99]	0.66 [4.11]	0.33 [3.63]	0.35 [3.82]	0.30 [3.26]	0.30 [3.24]	0.19 [2.23]	0.21 [2.42]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDI Quintiles					EDI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	384	385	385	379	358	37	39	34	30	24	
	(2)	107	108	108	107	105	57	58	57	57	55	
	(3)	78	78	78	78	78	100	99	101	100	99	
	(4)	65	65	65	65	65	213	211	218	213	214	
(5)	60	59	60	60	59	1201	1696	1936	1888	1508		

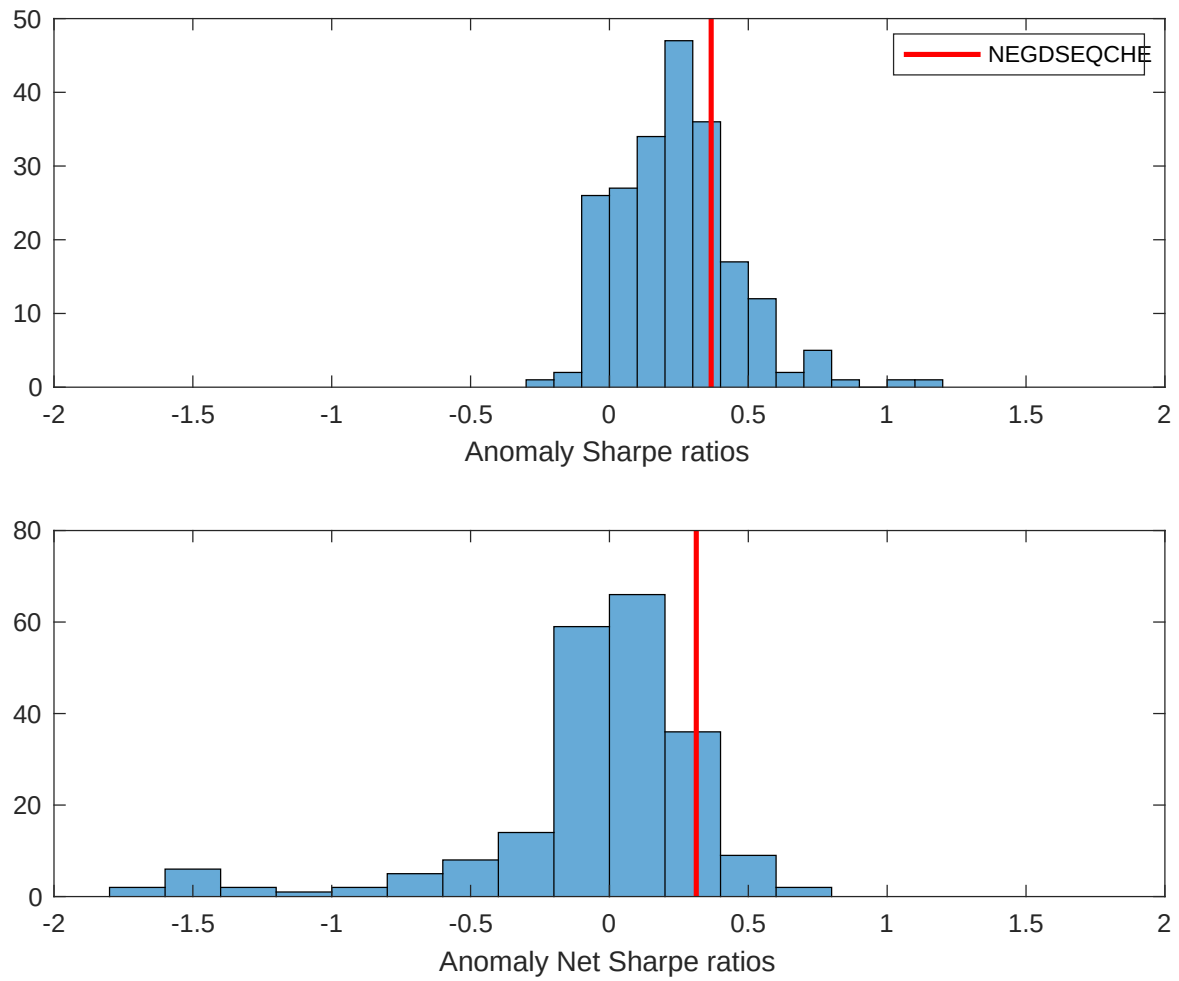


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

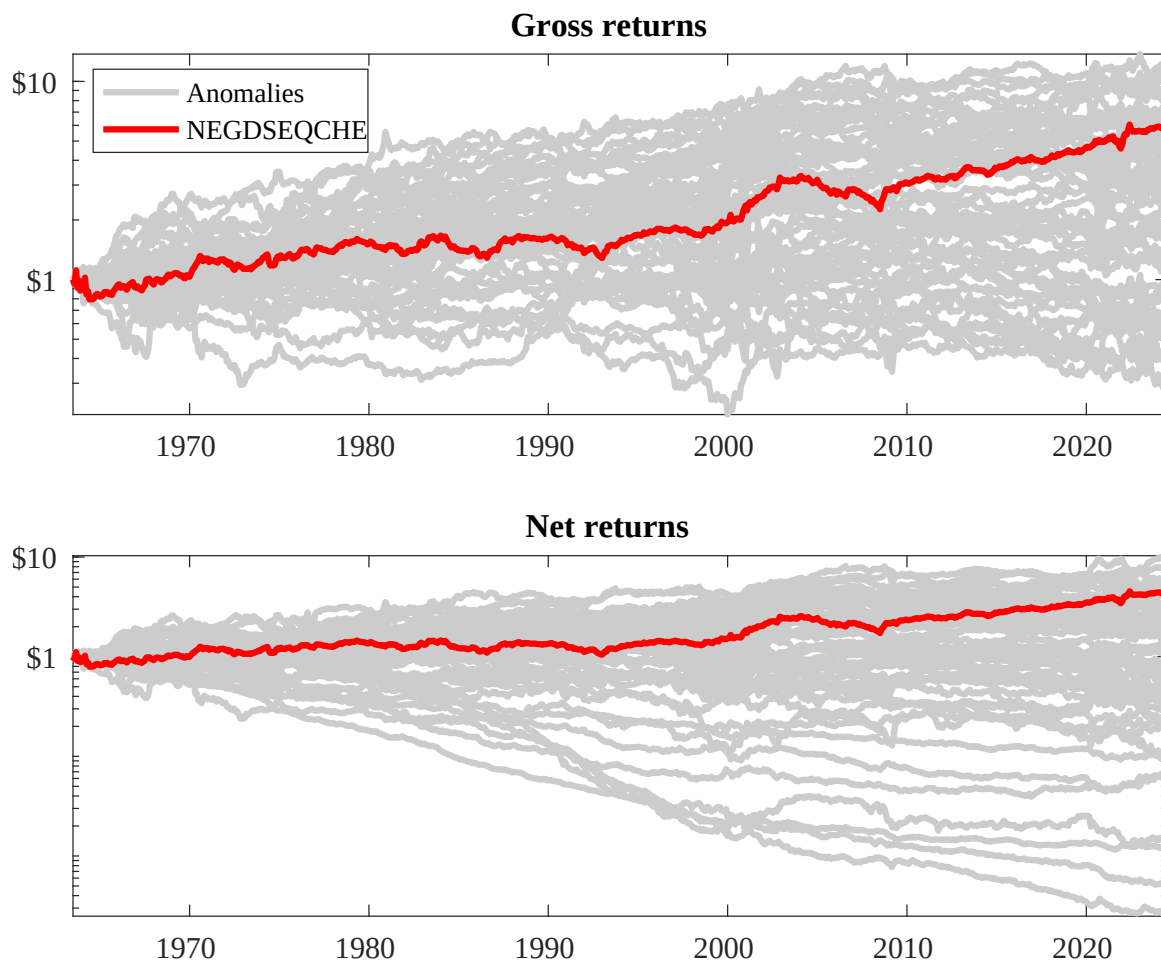
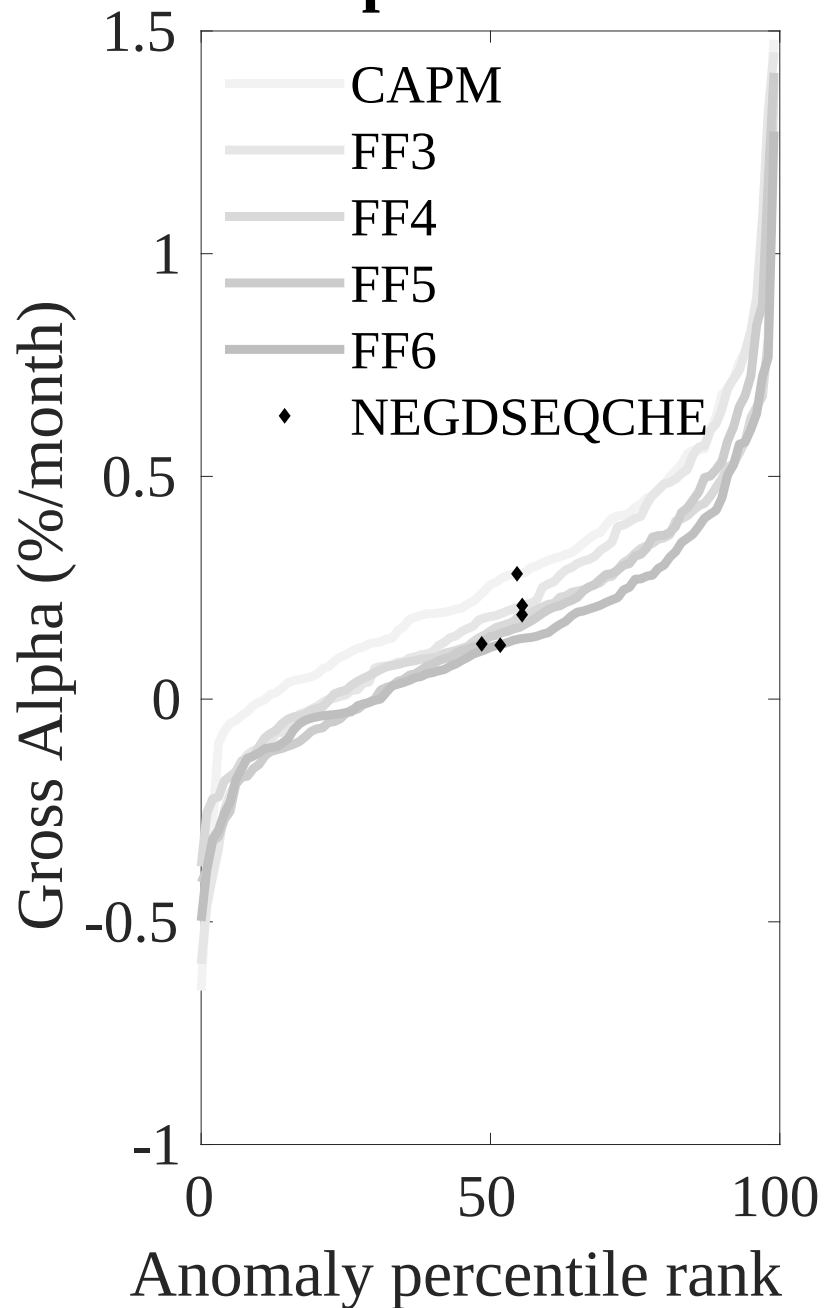


Figure 3: Dollar invested.

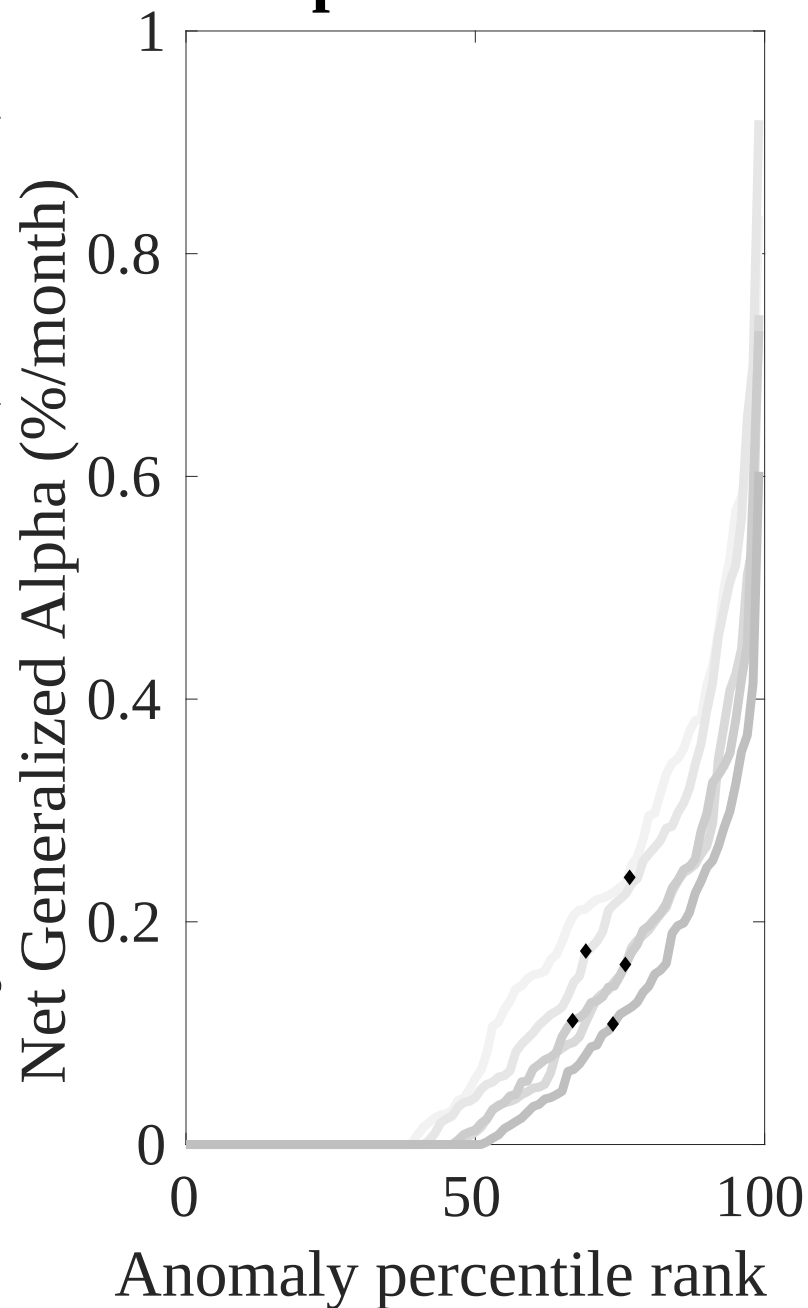
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



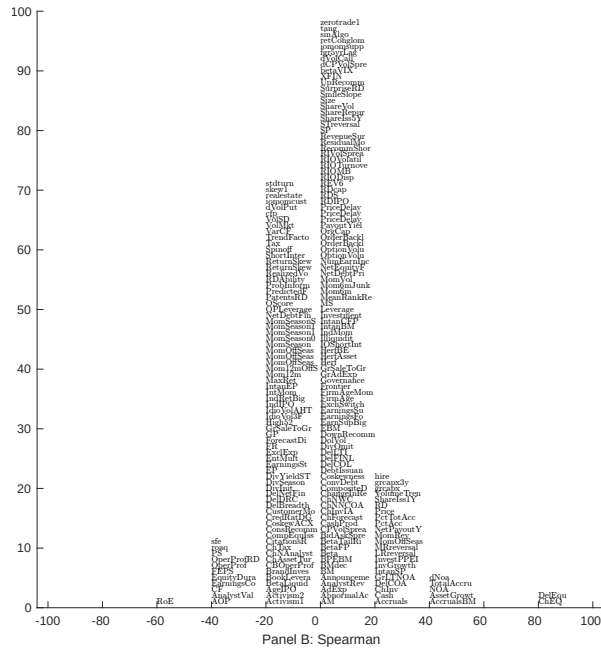
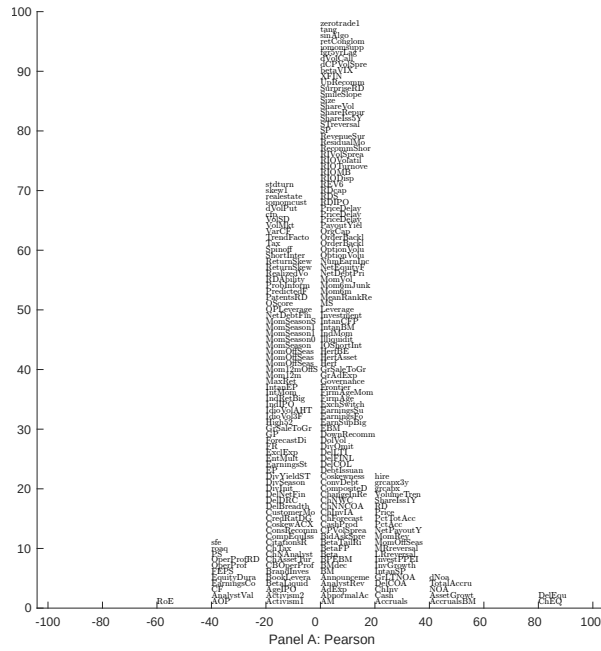


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

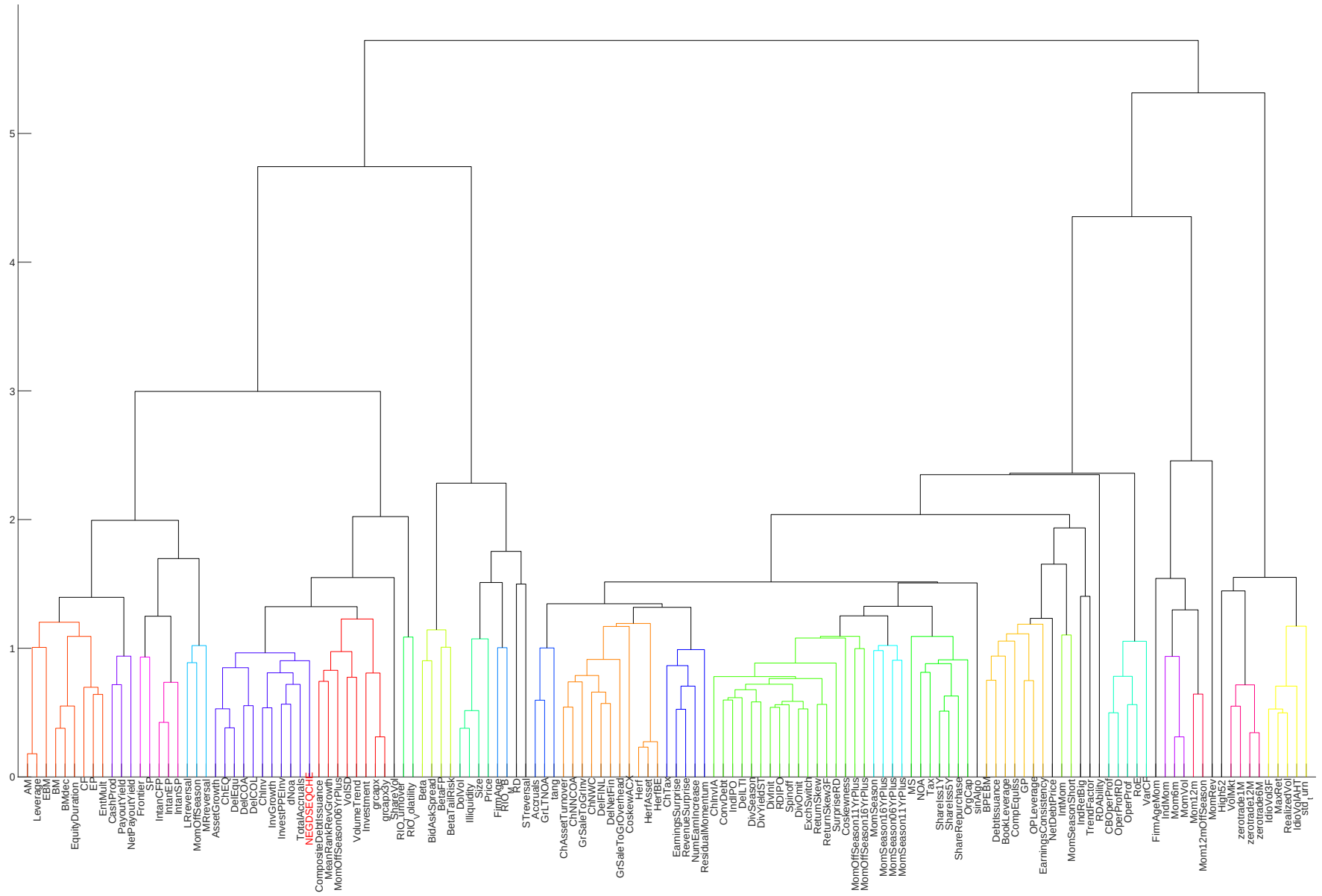


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

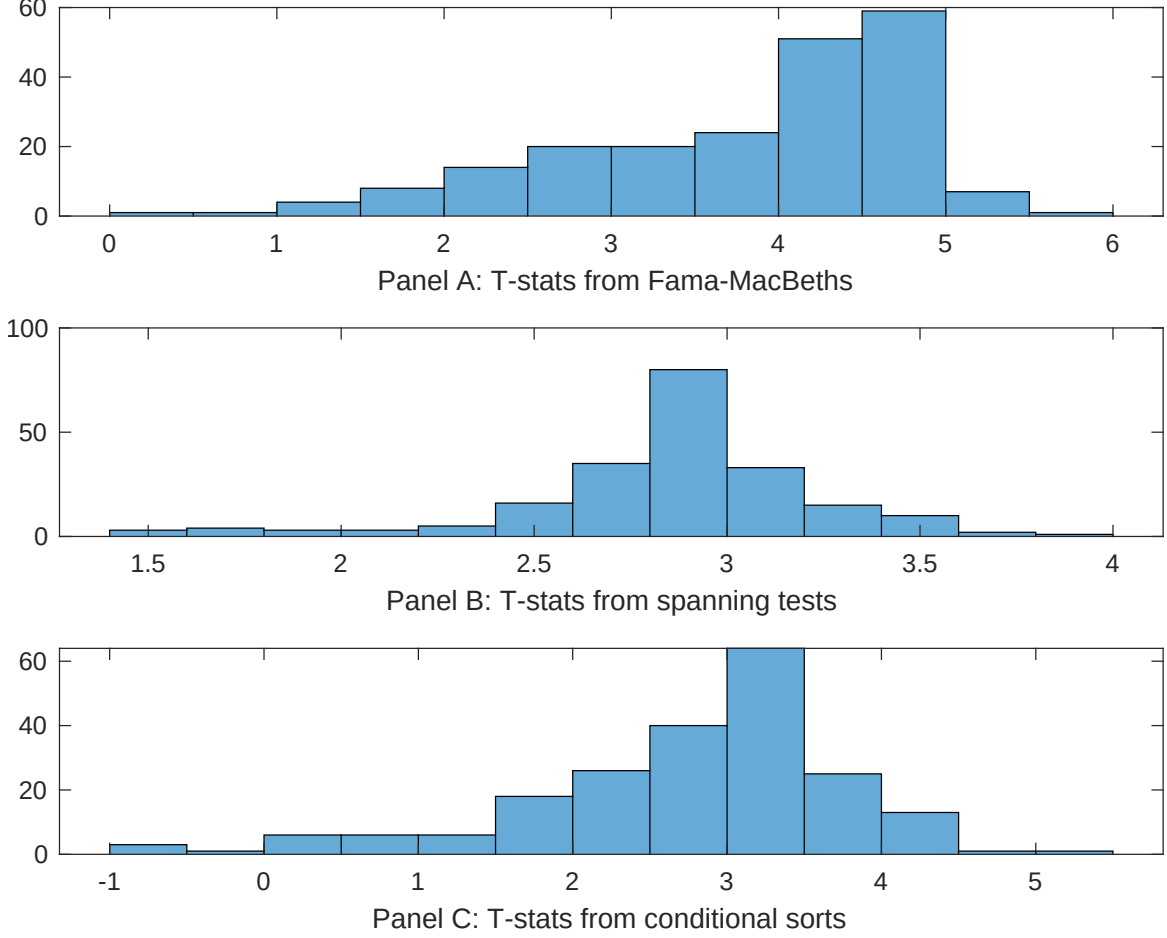


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDI}EDI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.67]	0.17 [7.39]	0.13 [6.10]	0.13 [5.94]	0.12 [5.40]	0.11 [3.92]	0.12 [5.05]
EDI	0.14 [3.15]	0.16 [3.60]	0.10 [2.20]	0.13 [2.52]	0.24 [4.43]	0.12 [3.34]	0.36 [1.15]
Anomaly 1	0.14 [4.00]						0.62 [1.50]
Anomaly 2		0.44 [4.31]					0.12 [0.13]
Anomaly 3			0.99 [9.29]				0.48 [3.73]
Anomaly 4				0.13 [9.91]			0.41 [2.36]
Anomaly 5					0.34 [1.82]		-0.25 [-1.02]
Anomaly 6						0.13 [3.11]	0.59 [1.37]
# months	708	708	708	708	708	432	432
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [1.48]	0.10 [1.15]	0.12 [1.36]	0.08 [0.87]	0.09 [1.03]	0.26 [2.87]	0.23 [2.82]
Anomaly 1	41.89 [9.60]						10.17 [1.24]
Anomaly 2		47.92 [10.11]					47.47 [6.50]
Anomaly 3			23.29 [4.24]				-21.05 [-3.59]
Anomaly 4				29.57 [5.77]			13.79 [2.67]
Anomaly 5					20.50 [4.85]		-1.40 [-0.27]
Anomaly 6						19.94 [4.95]	11.17 [2.70]
mkt	5.67 [2.70]	8.01 [3.84]	6.47 [2.94]	6.58 [3.02]	5.73 [2.61]	-0.55 [-0.24]	1.25 [0.61]
smb	5.49 [1.81]	4.42 [1.46]	4.16 [1.30]	6.34 [2.02]	6.11 [1.93]	4.84 [1.48]	2.99 [1.01]
hml	-14.74 [-3.60]	-15.02 [-3.69]	-9.91 [-2.33]	-11.46 [-2.71]	-8.96 [-2.12]	-5.97 [-1.49]	-9.73 [-2.61]
rmw	-5.61 [-1.37]	-7.19 [-1.77]	-9.78 [-2.28]	-9.20 [-2.17]	-6.21 [-1.44]	-13.58 [-3.25]	-17.18 [-4.52]
cma	20.34 [2.75]	16.08 [2.15]	34.42 [3.78]	40.47 [5.57]	52.23 [7.95]	51.70 [8.67]	15.36 [1.99]
umd	1.46 [0.70]	-0.38 [-0.18]	1.21 [0.55]	-0.13 [-0.06]	0.98 [0.45]	0.61 [0.30]	-1.85 [-0.99]
# months	731	731	731	731	731	438	438
$\bar{R}^2(\%)$	27	28	19	21	20	31	46

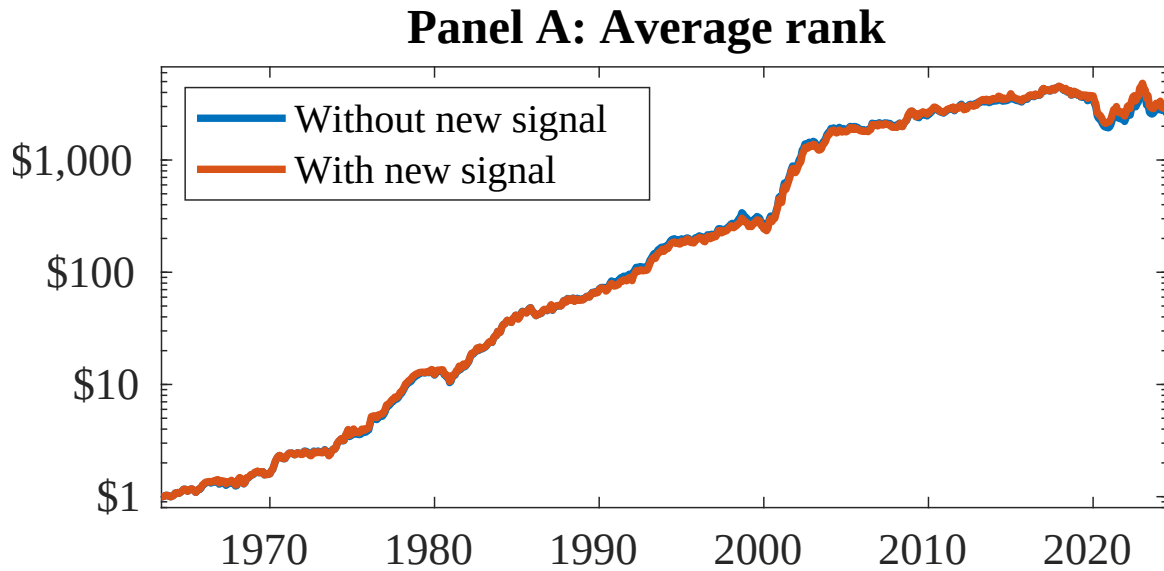


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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